CAPÍTULO 1
1.4 Exercícios – pág. 18 – Continuação
(de 13 até 17- final)

12. Escrever a função que representa o parabolóide circular das figuras 1.36, 1.37 e 1.38.

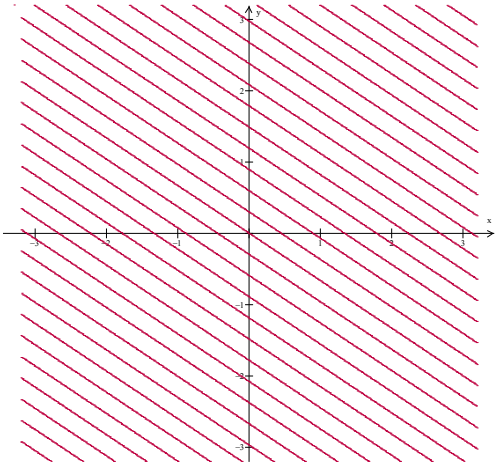
Para a figura 1.36 temos um parabolóide com concavidade para cima e vértice na origem. Como *z(0,3)=4*, usando a equação $z = c\left(x^2 + y^2\right)$, obtemos $c = \frac{4}{3}$. Logo, $z = \frac{4}{9}\left(x^2 + y^2\right)$.

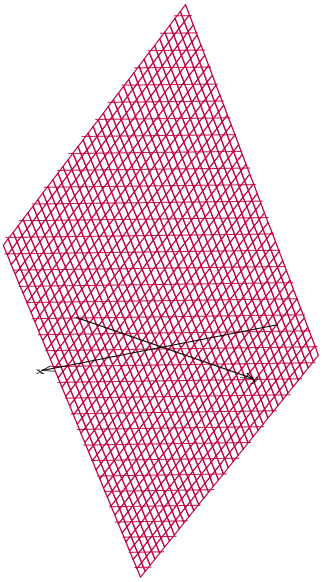
Para a figura 1.37 temos $z = 4 - \frac{4}{9}\left(x^2 + y^2\right)$, pois o vértice do parabolóide está no ponto $(0,0,4)$ e o mesmo tem a concavidade para baixo. Usando a equação $z = 4 - c\left(x^2 + y^2\right)$, obtemos $c = \frac{4}{9}$.

Para a figura 1.38 temos $z = \frac{3}{2} + \frac{5}{18}\left(x^2 + y^2\right)$, pois o parabolóide tem o vértice no ponto $(0,0,3/2)$; está virado para cima e tem curva de nível para $z = 4$ igual a $x^2 + y^2 = 9$. Usando as equações $z = \frac{3}{2} + c\left(x^2 + y^2\right)$ e *z(0,3)=4*, obtém-se $c = \frac{5}{18}$.

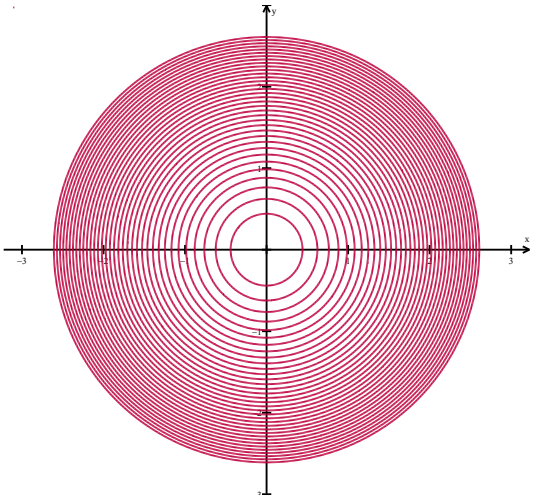
13. Desenhar algumas curvas de nível e esboçar o gráfico:

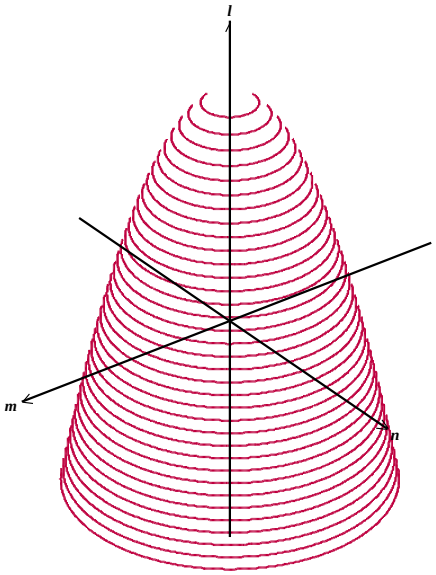
a) $z = 3 - 2x - 3y$



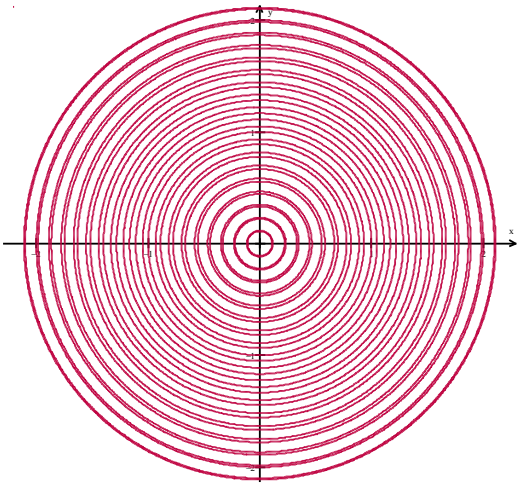


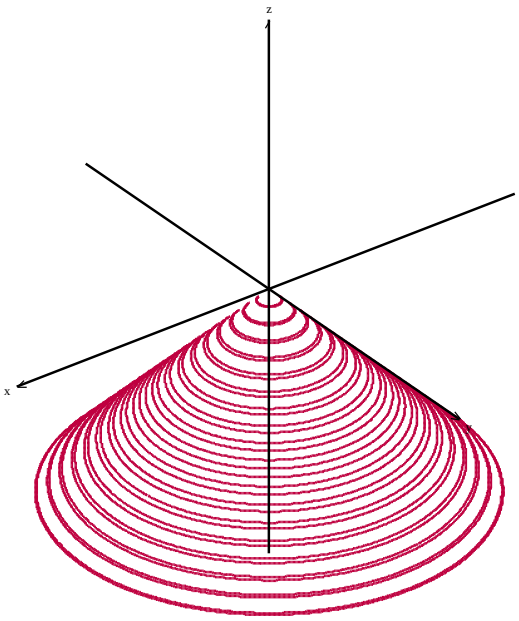
b) $l = 4 - m^2 - n^2$



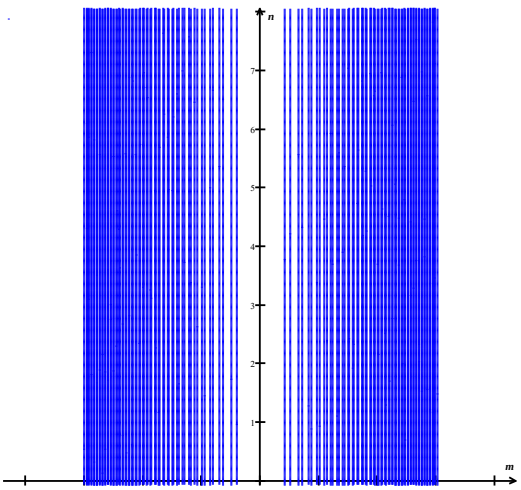


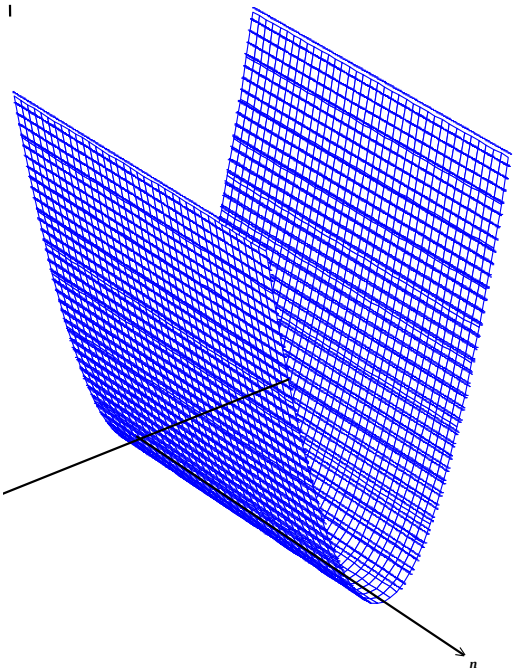
c) $z = -\sqrt{x^2 + y^2}$



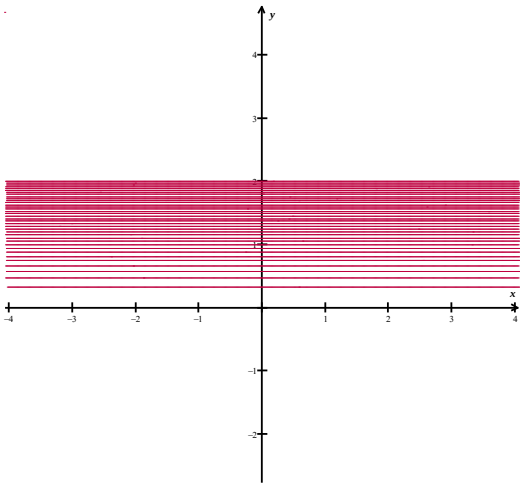


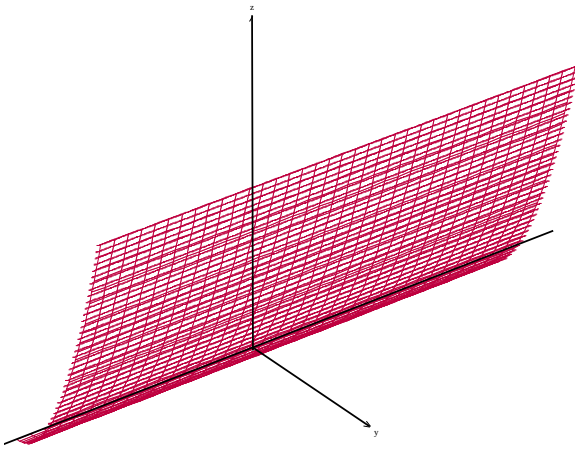
d) $l = m^2, 0 \leq n \leq 8$



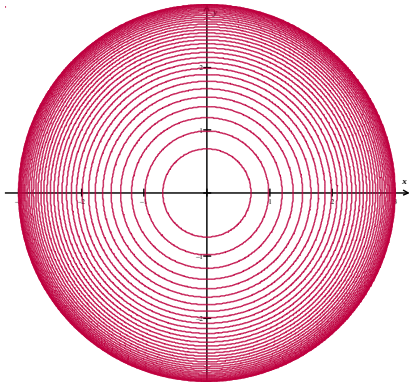


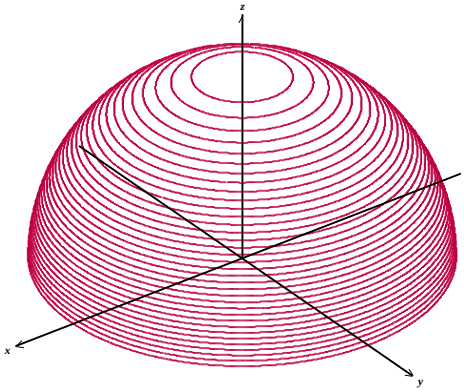
e) $z = y^2, -4 \leq x \leq 4, 0 \leq y \leq 2$



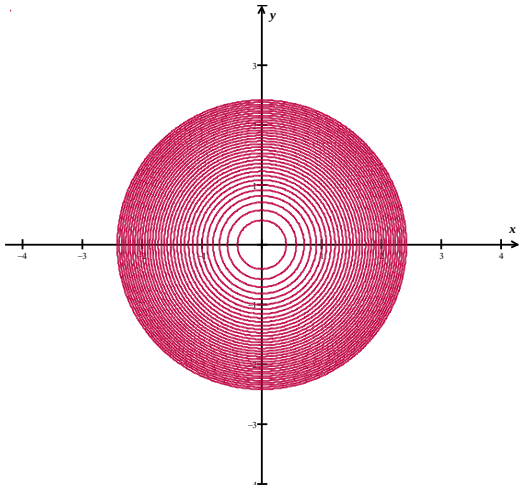


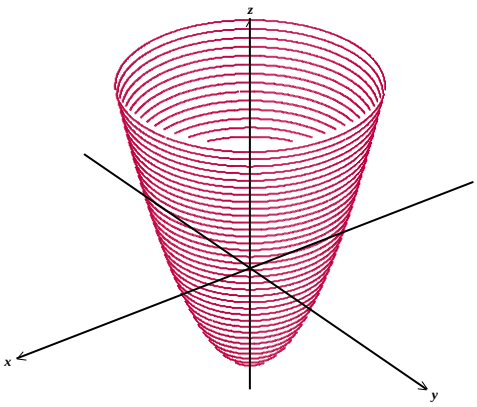
f) $z = \sqrt{9 - x^2 - y^2}$



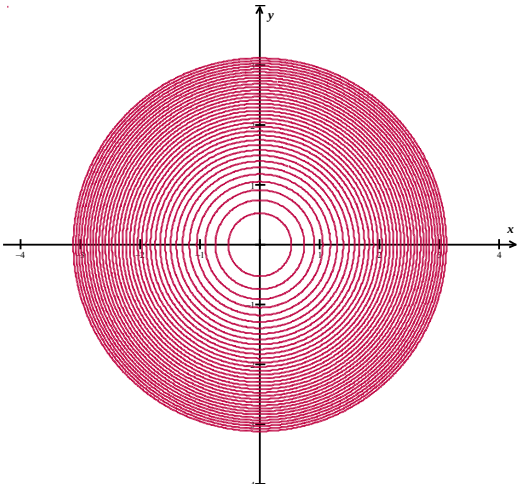


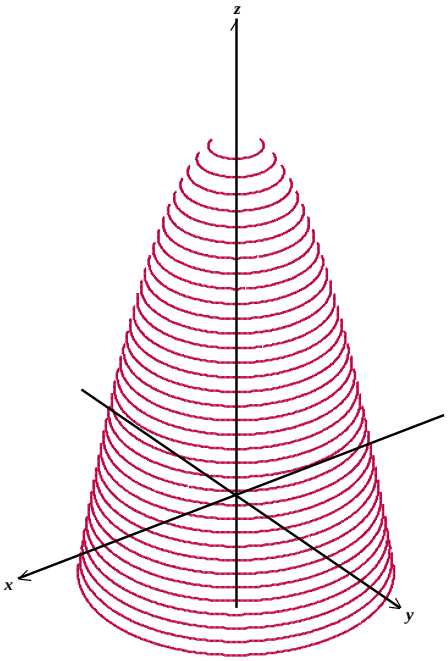
g) $z = x^2 + y^2 - 2$



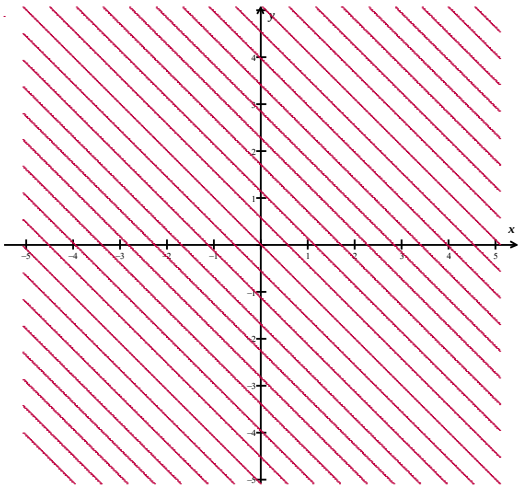


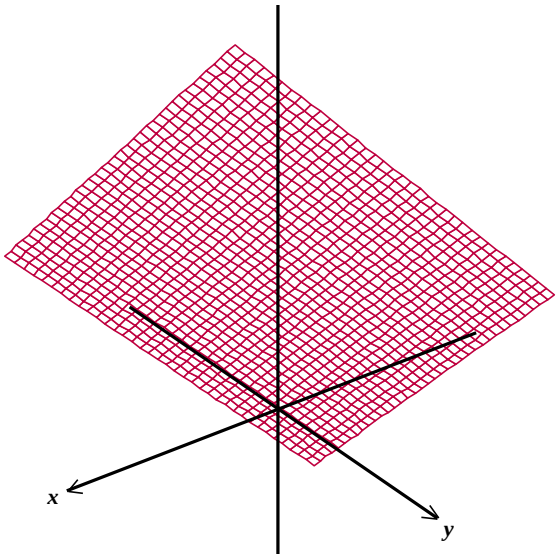
h) $z = 8 - x^2 - y^2$





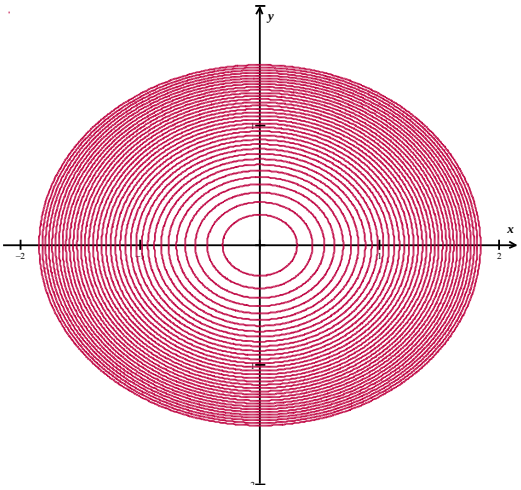
i) $f(x,y) = x + y + 4$

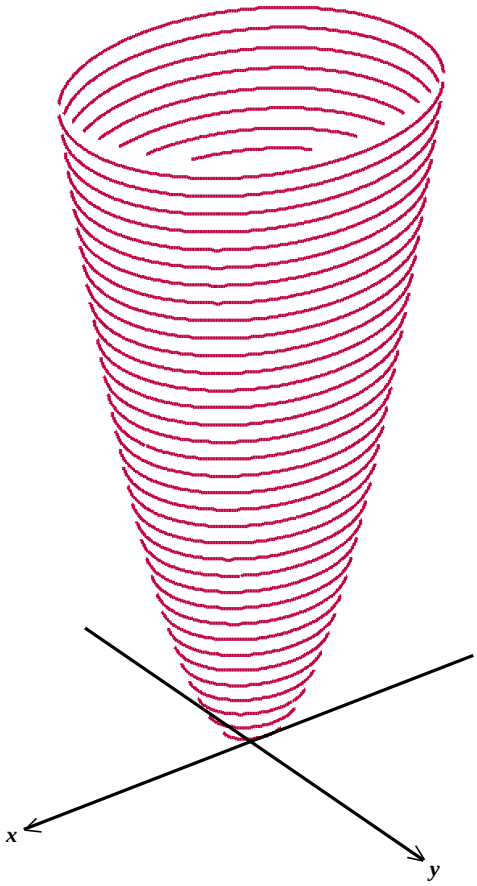




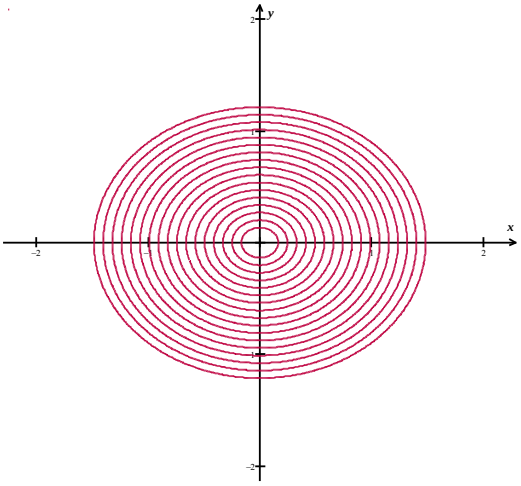
j) $z = 2x^2 + 3y^2$

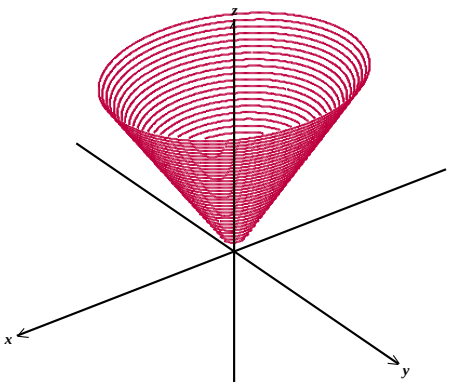
I



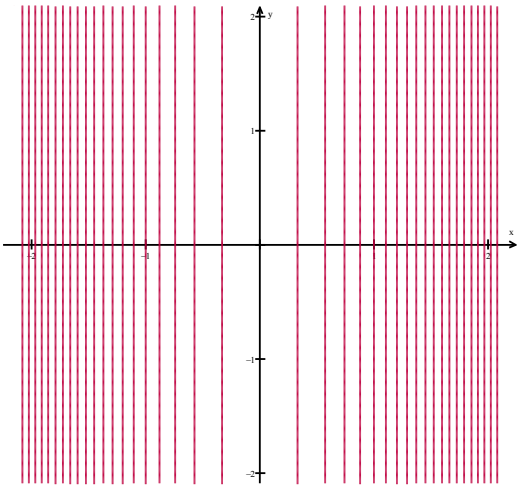


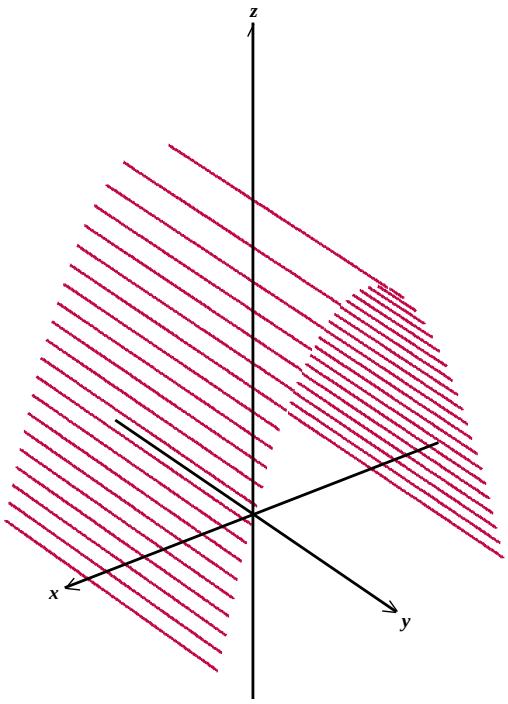
k) $z = \sqrt{2x^2 + 3y^2}$



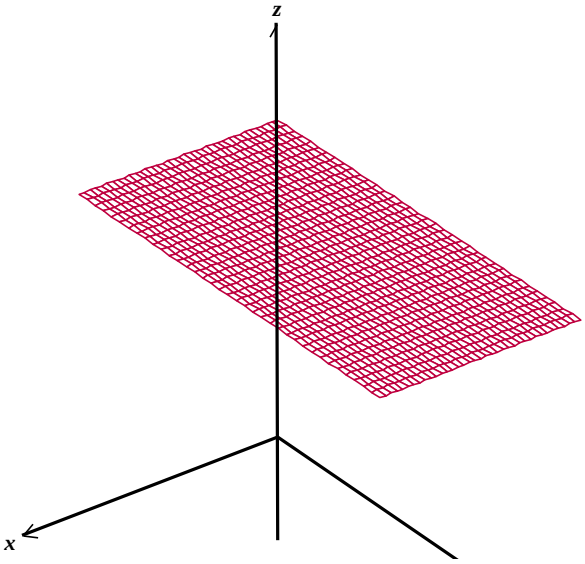


l) $z = 4 - x^2$

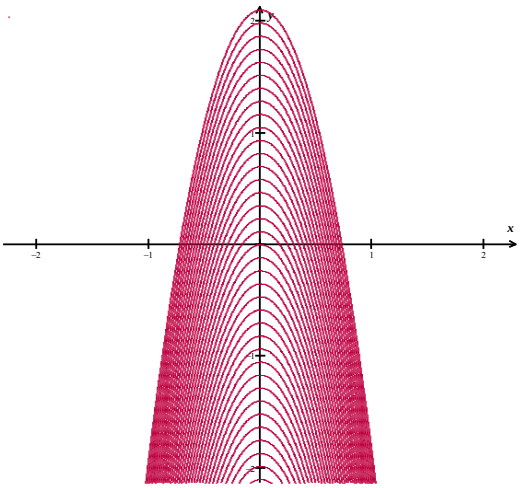


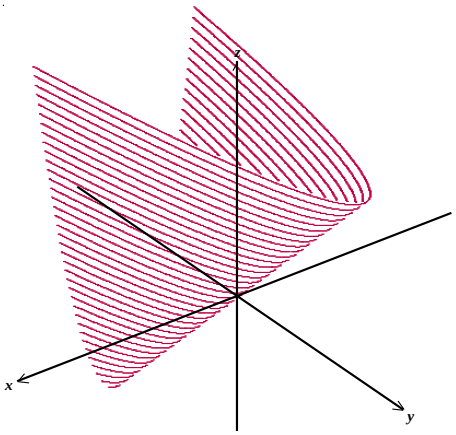


m) $z = 3, 0 \leq x \leq 2 \quad e \quad 0 \leq y \leq 4$

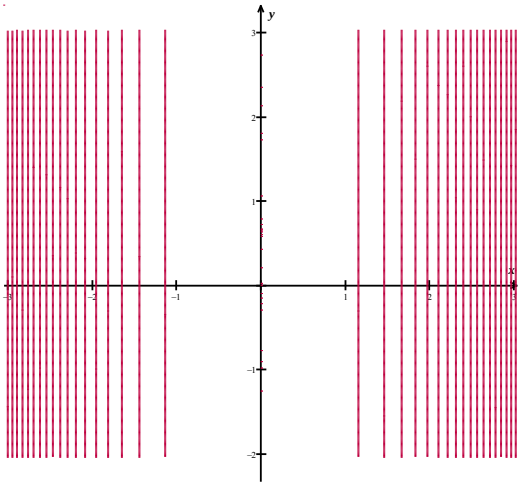


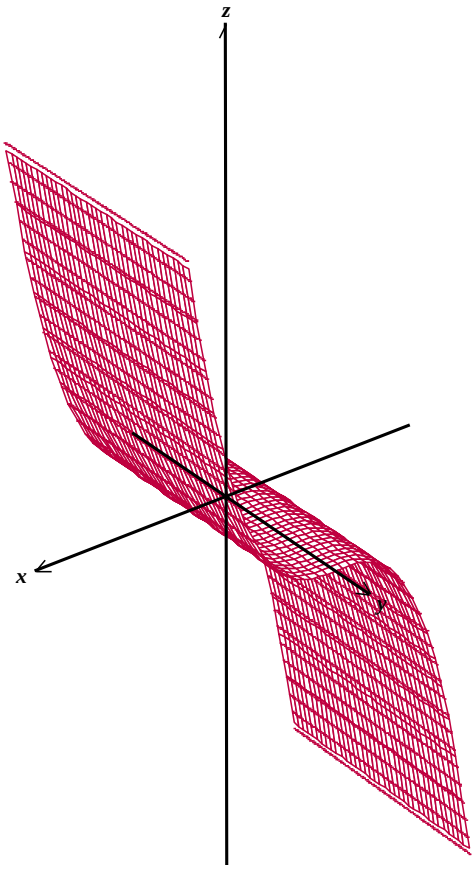
n) $z = 4x^2 + y$



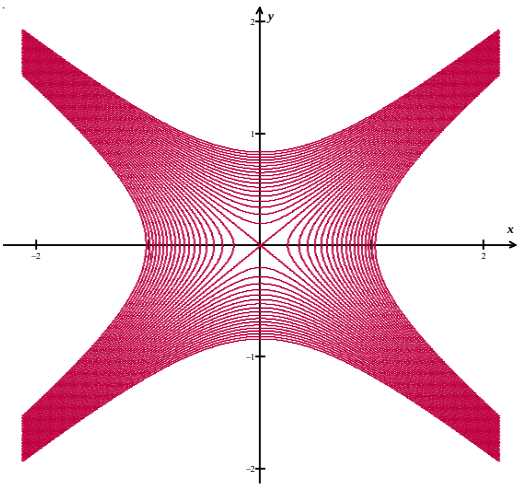


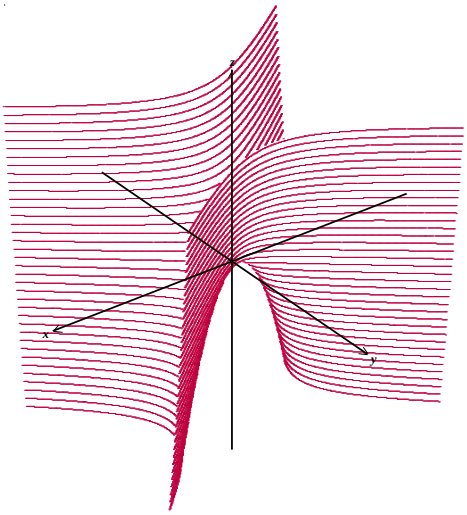
o) $z = \frac{1}{4}x^3$



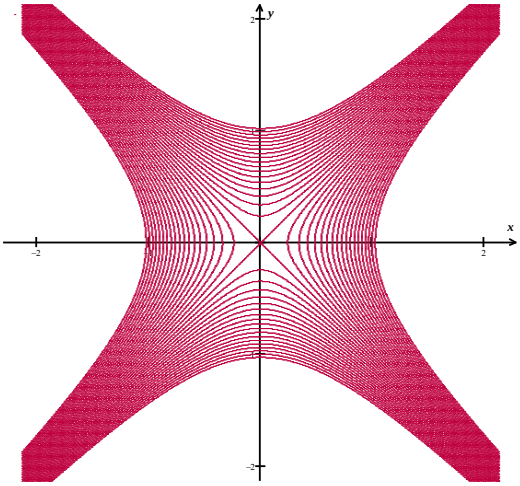


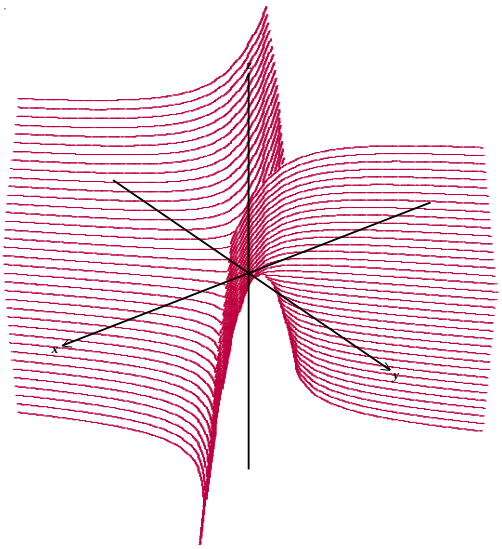
p) $z = 2x^2 - 3y^2$





q) $z = 3y^2 - 2x^2$





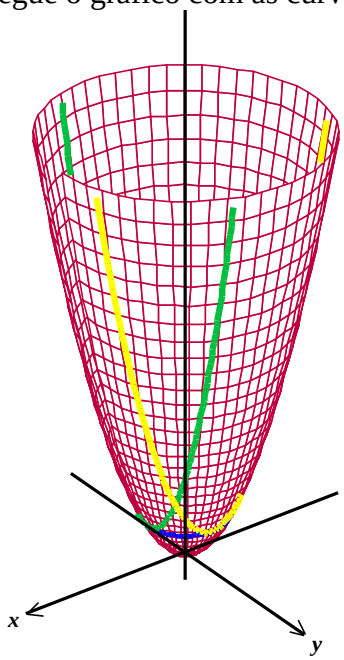
14. Encontrar a curva de intersecção do gráfico da função dada com os planos dados, representando graficamente:

a) $z = x^2 + y^2$ com os planos *z=1*; *x=1*;*y=1*.

As intersecções são dadas por:

$$\begin{cases} x^2+y^2=1 \\ z=1 \end{cases} \quad \begin{cases} z=1+y^2 \\ x=1 \end{cases} \quad \begin{cases} z=1+x^2 \\ y=1 \end{cases}$$

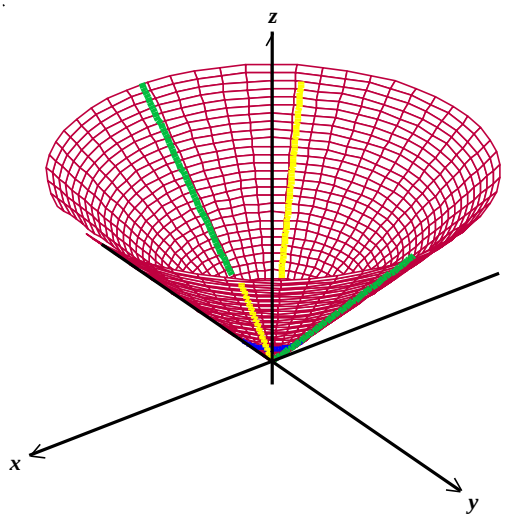
Segue o gráfico com as curvas assinaladas em cores azul, verde e amarela.



b) $z=\sqrt{x^2+y^2}$ com os planos *z=1*; *x=0*;*y=x*.
As intersecções são dadas por:

$$\begin{cases} x^2+y^2=1 \\ z=1 \end{cases} \quad \begin{cases} z=|y| \\ x=0 \end{cases} \quad \begin{cases} z=\sqrt{2}\,|x| \\ y=x \end{cases}$$

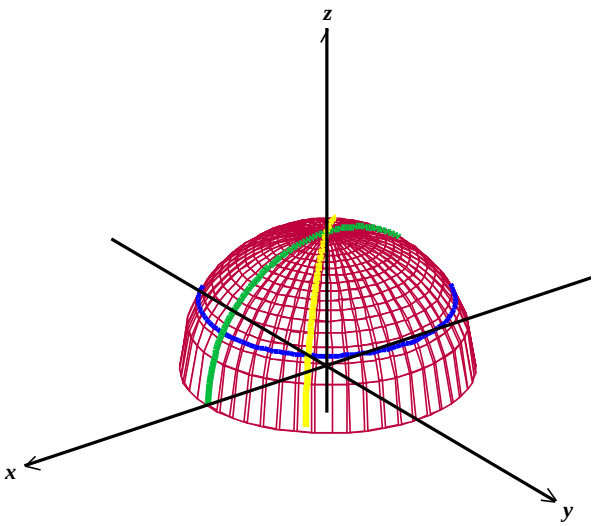
Segue o gráfico com as curvas assinaladas em cores azul, verde e amarela.



c) $z=\sqrt{4-x^2-y^2}$ com os planos *z=1*; *y=0*;*y=x*.
As intersecções são dadas por:

$$\begin{cases} x^2+y^2=3 \\ z=1 \end{cases} \quad \begin{cases} z=\sqrt{4-x^2} \\ y=0 \end{cases} \quad \begin{cases} z=\sqrt{4-2x^2} \\ y=x \end{cases}$$

Segue o gráfico com as curvas assinaladas em cores azul, verde e amarela.

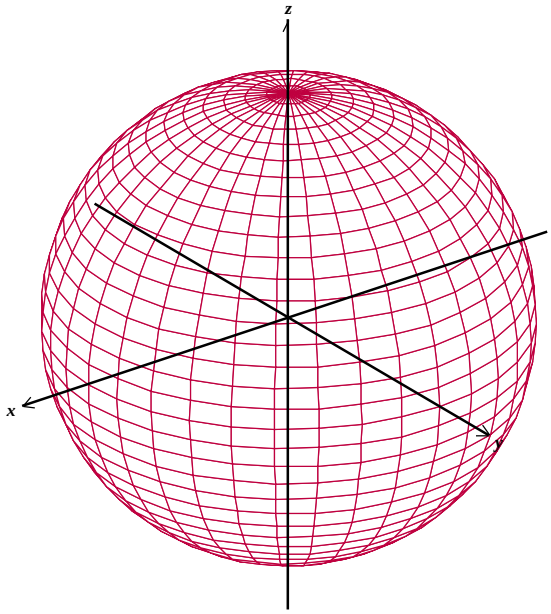


15. Esboçar o gráfico das superfícies de nível S_k correspondentes aos valores de k dados:

a) $w=x^2+y^2+z^2$; *k=0,1,4,9.*

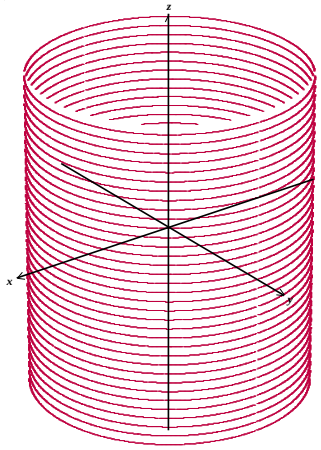
Para k=0, temos $x^2+y^2+z^2=0$, que é uma superfície degenerada, pois temos apenas o ponto (0,0,0).

Para k=1,4,9 temos esferas centradas na origem de raio 1, 2 e 3 respectivamente.
Na figura que segue mostramos a esfera de raio 2.



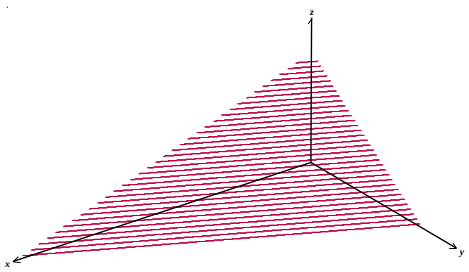
b) $w = x^2 + y^2$; *k = 4,16,25.*

Neste caso vamos ter cilindros verticais infinitos de raio 2, 4,e 5
($x^2 + y^2 = 4$; $x^2 + y^2 = 16$; $x^2 + y^2 = 25$).
Na figura que segue, mostramos o cilindro de raio 4, delimitado inferiormente e superiormente.



c) $w = x + 2y + 3z$; *k=1,2,3.*

Neste caso, vamos ter planos $x + 2y + 3z = 1$; $x + 2y + 3z = 2$; $x + 2y + 3z = 3$
A figura mostra a parte do plano $x + 2y + 3z = 1$ do primeiro octante.



16. Sabendo que a função $T(x, y, z) = 30 - \left(x^2 + \frac{y^2}{4} + \frac{z^2}{9} \right)$ representa a temperatura nos pontos da região do espaço delimitada pelo elipsóide $x^2 + \frac{y^2}{4} + \frac{z^2}{9} = 1$, pergunta-se:

a) Em que ponto a temperatura é a mais alta possível?

Temos a maior temperatura na origem $(0,0,0)$ quando $x^2 + \frac{y^2}{4} + \frac{z^2}{9} = 0$. Nesse caso, a temperatura mais alta possível assume o valor 30 unidades de temperatura.

b) Se uma partícula se afasta da origem, deslocando-se sobre o eixo positivo dos x, sofrerá aumento ou diminuição de temperatura?
Diminuição.

c) Em que pontos a temperatura é a mais baixa possível?
Na casca da superfície do elipsóide $x^2 + \frac{y^2}{4} + \frac{z^2}{9} = 1$.

17. Fazer um esboço de algumas superfícies de nível da função $w = \sqrt{x^2 + y^2 + z^2}$. O que ocorre com os valores da função ao longo de semi-retas que partem da origem?

As superfícies de nível são esferas centradas na origem. Os valores da função crescem à medida que nos afastamos da origem.